Power BI Data Cleaning and ETL

A Blueprint for the Power Query Editor Workbench

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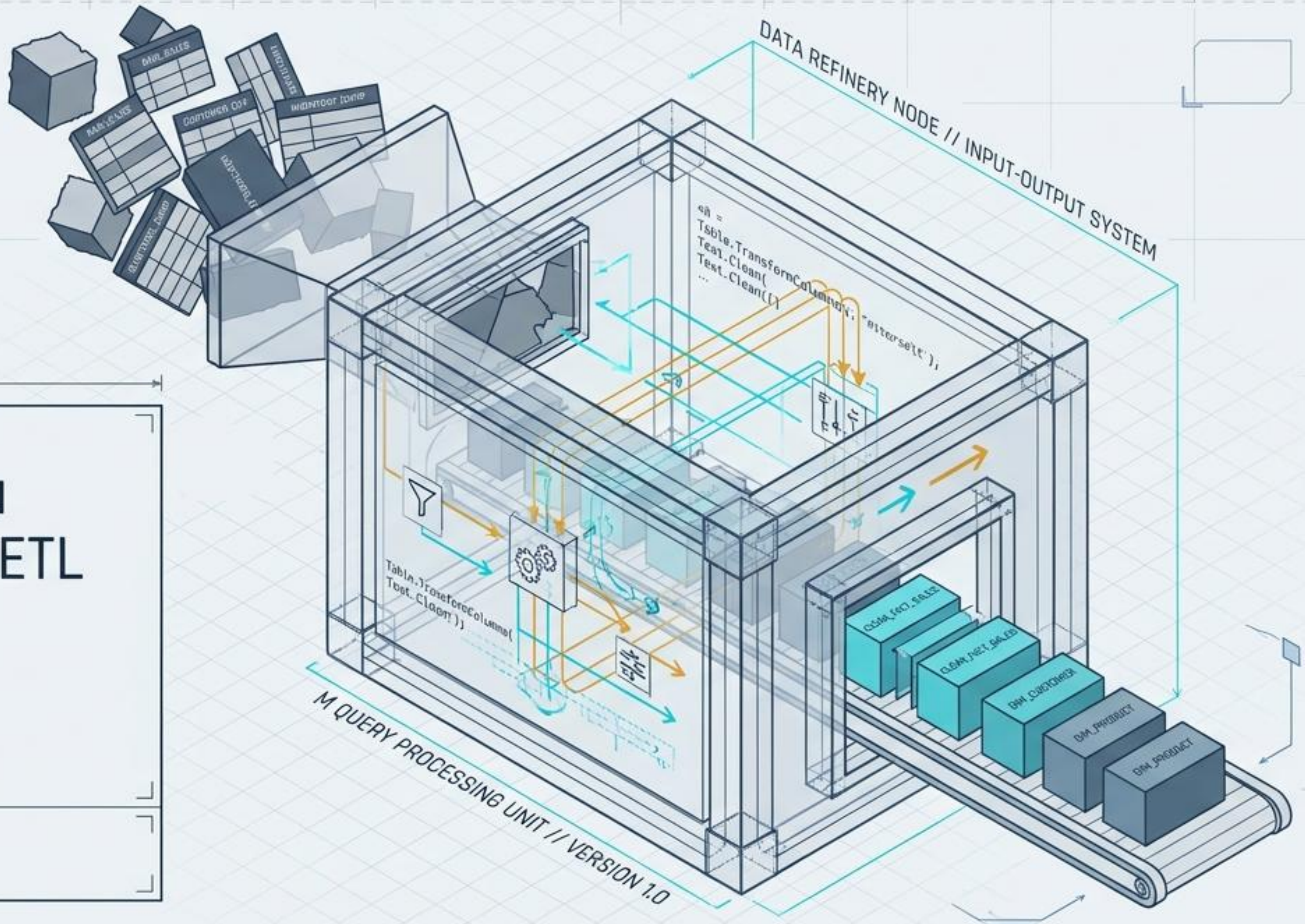
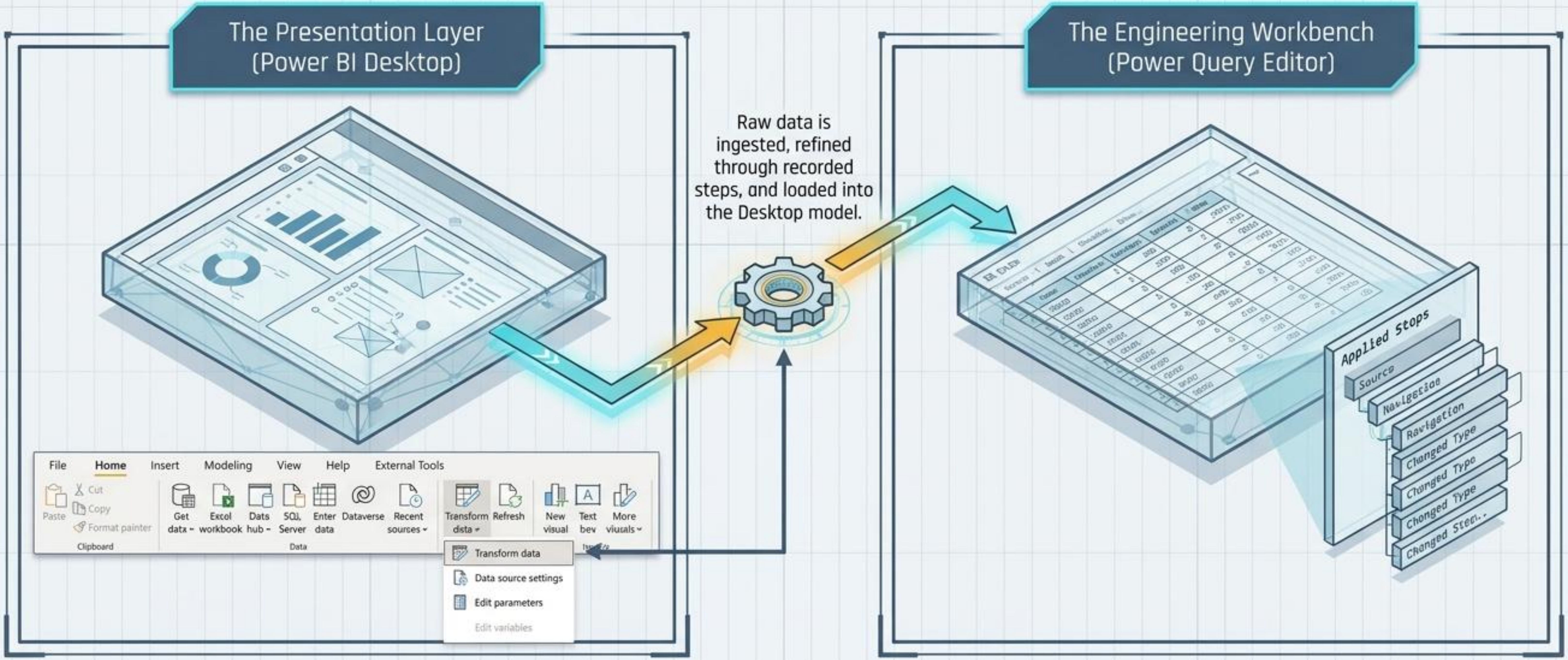


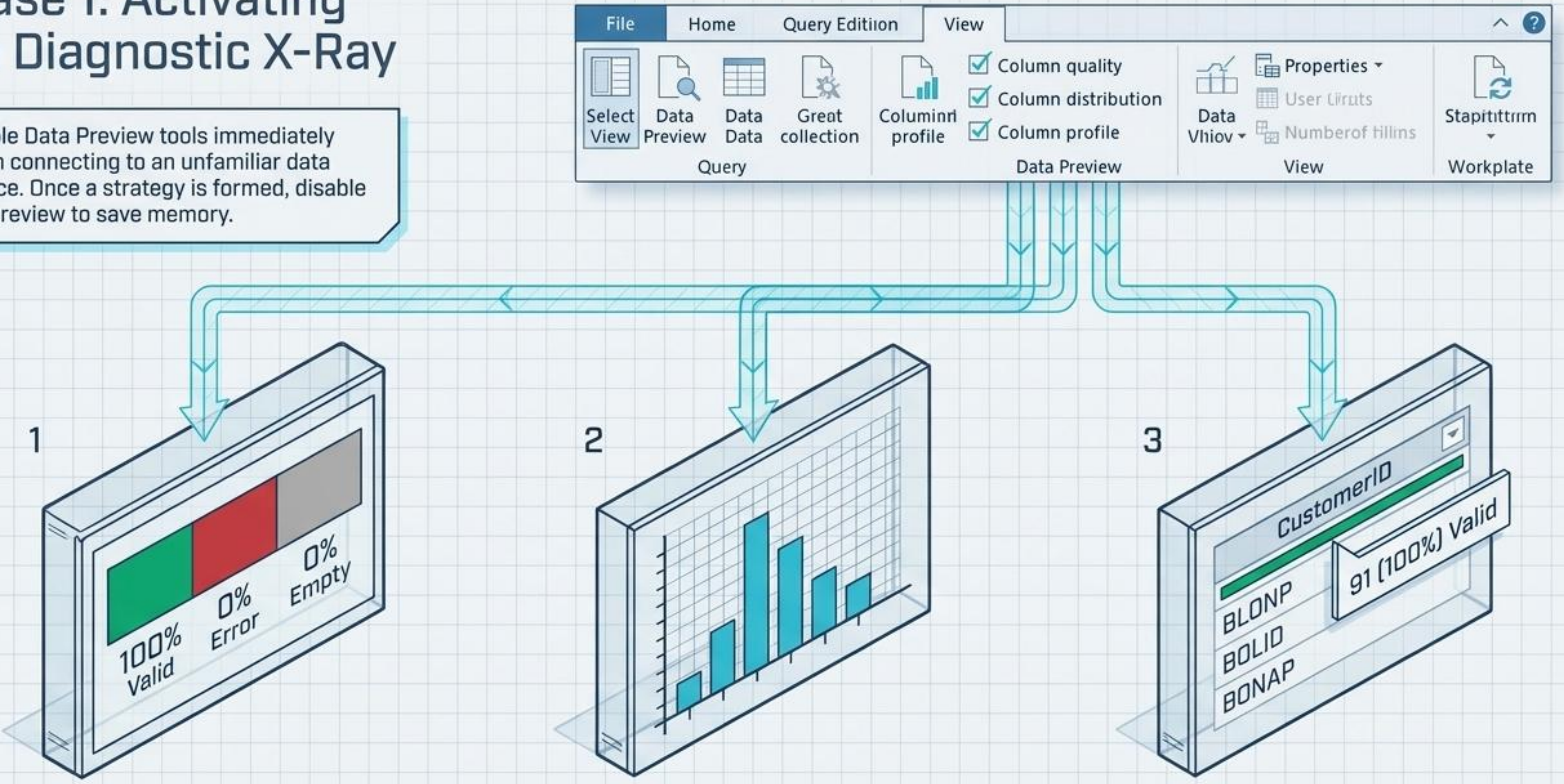
The Dual-Engine Architecture

The Power Query Editor (PQE) is not a standalone application. It is a specialized, side-by-side engineering environment accessed directly via Power BI Desktop.



Phase 1: Activating the Diagnostic X-Ray

Enable Data Preview tools immediately when connecting to an unfamiliar data source. Once a strategy is formed, disable full preview to save memory.



Column Quality Meter

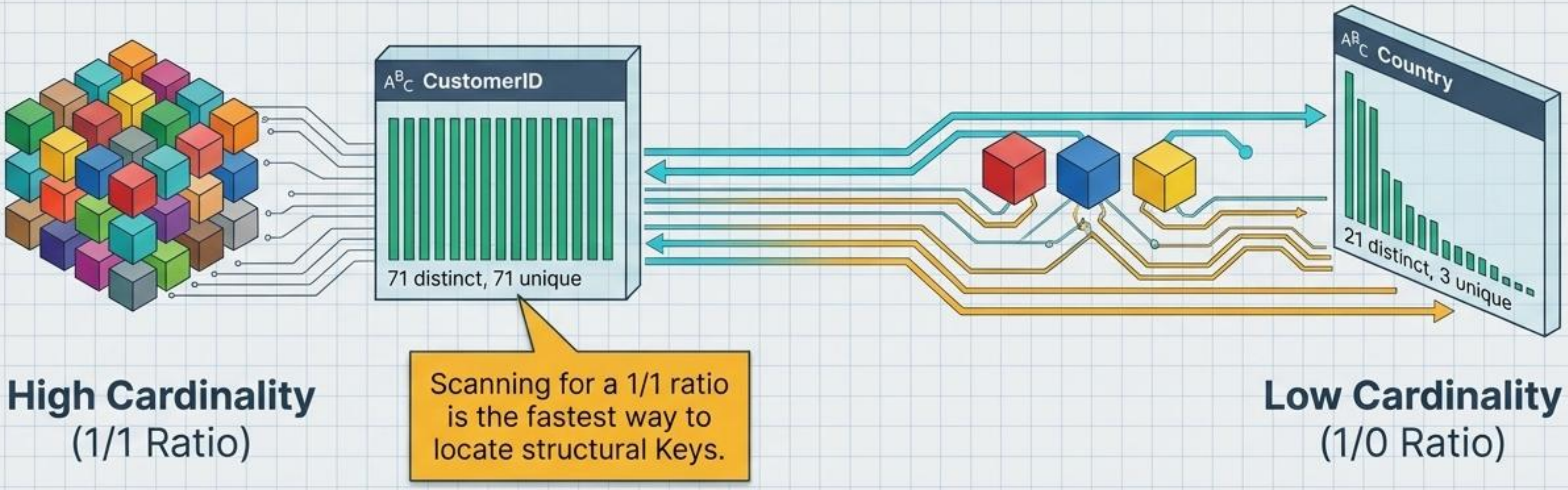
Column Distribution Meter

The Shortcut Bar

The Cardinality Spectrum

Distinct: The number of different values present in a column.

Unique: The number of values that appear exactly once and are never repeated.



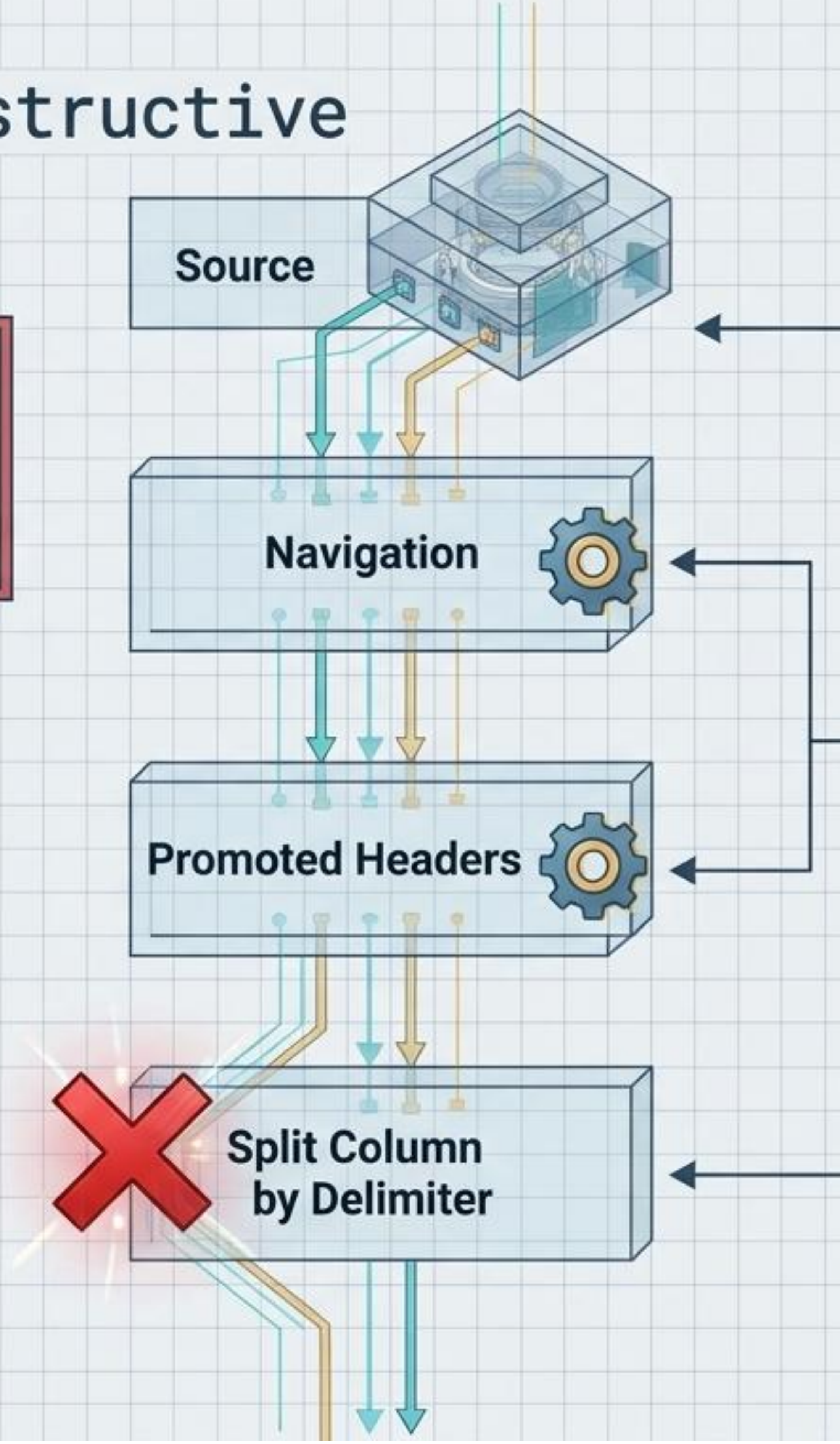
Anatomy of Keys: The 1/1 Identifiers

Rule: Every dataset requires a unique identifier (Primary Key) to establish relationships. Simplify these keys to the minimum necessary characters.

Origin	Location: Primary	Location: Foreign
Native	Q1 **Native Primary Key** The 1/1 high-cardinality identifier in its home table, sourced from business logic (e.g., HR ID 'XYZ123'). Vulnerable to taxonomy changes.	Q3 **Native Foreign Key** The corresponding business-logic column in a related table, used to form data relationships.
Surrogate	Q2 **Surrogate Primary Key** The 1/1 identifier in its home table, injected by a database designer (e.g., '123456'). Immutable and independent of business logic changes.	Q4 **Surrogate Foreign Key** The corresponding database-logic column in a related table, used to form data relationships.

Phase 2: The Non-Destructive Step Recorder

The Paradigm Shift: The Power Query Editor contains no traditional 'Undo' button in the ribbon. Mistakes are fixed chronologically.



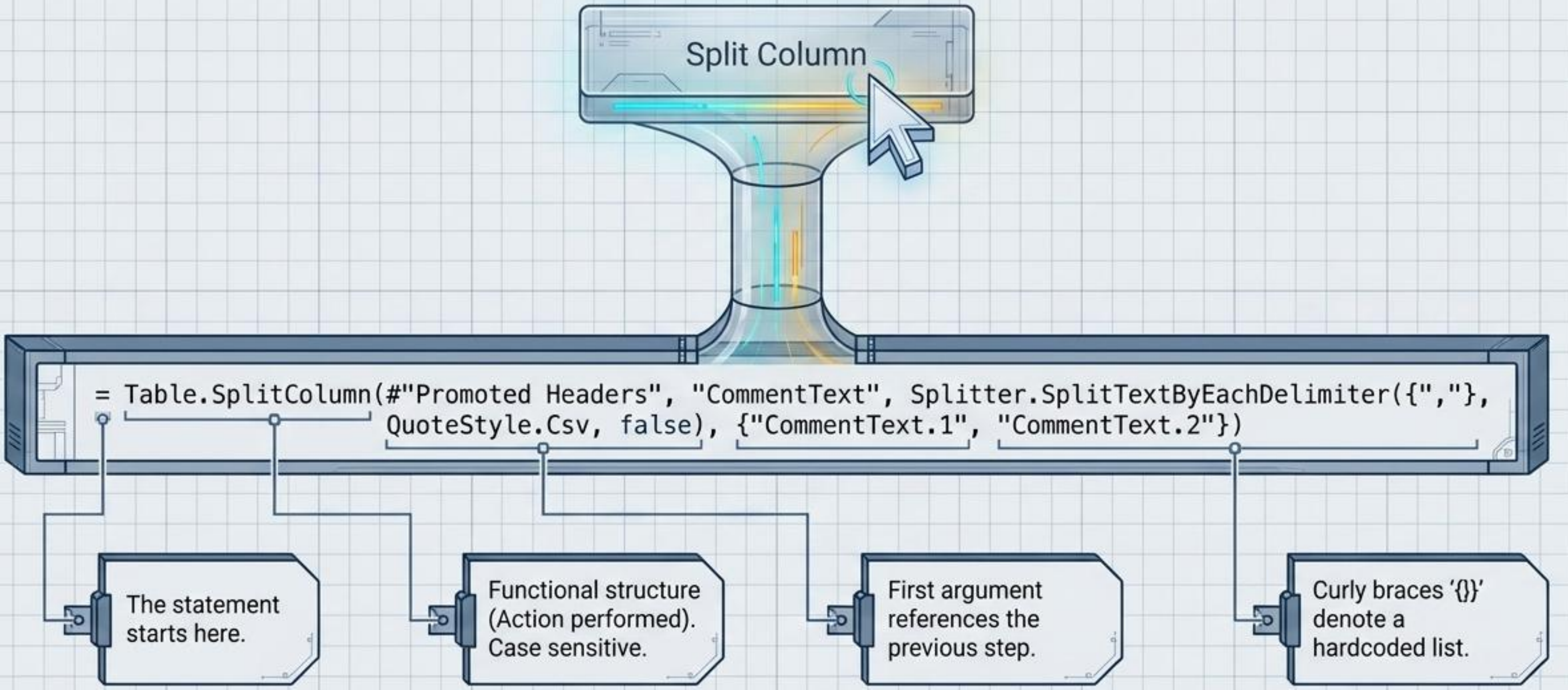
Mouse clicks are captured as individual sequential nodes called **Applied Steps.**

Target specific steps with the **gear icon** to modify parameters.

Use the red **'X'** to delete an action without destroying the surrounding pipeline.

Translating UI to the 'M' Language

Concept: M is a distinct, functional, case-sensitive language. The UI is just a graphical skin for functional programming.



Engineering a Panic-Proof Workbench

The Messy Workbench

Changed Type1
Removed Columns
Filtered Rows

The Documented Workbench

Source
Navigation
Changed Type
Promoted Headers
Removed UK and US

Manually rename steps to prevent panic-searching

Query Properties

Name: Customers

Description: Removes data for UK and US regions due to reporting requirements.

- [ ] Enable query properties
- [ ] Enable this query properties

Desktop Tooltip
Descriptions added here appear as tooltips in the presentation layer.

Playbook Rules

1. **Rename Filtered Steps:** Always manually rename steps where rows are excluded.
2. **Consolidate Renames:** Group column renames into a single step near the end.
3. **Use Single Quotes:** Table names with spaces must be wrapped in single quotes (e.g., 'Name Name2').
4. **Leverage 'All Properties':** Descriptions added here appear as tooltips in the presentation layer.

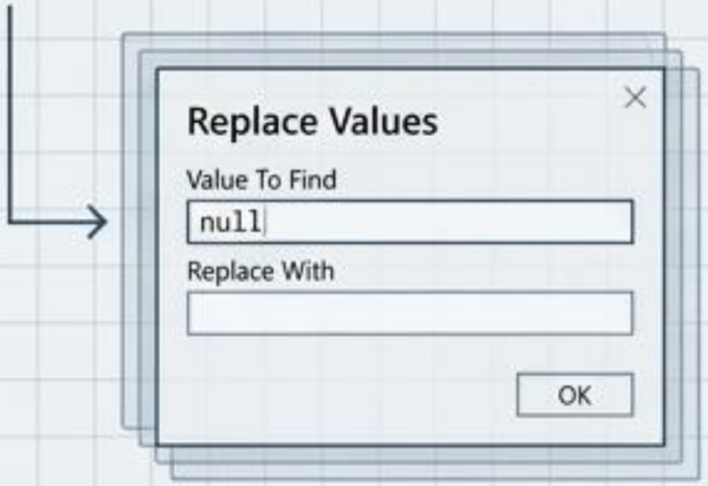
Phase 3: The Imperfection Triage Matrix

Concept: Power Query treats Nulls and Blanks as fundamentally different entities requiring specific formatting in the 'Replace Values' dialogue.

Null

System-recognized empty state.

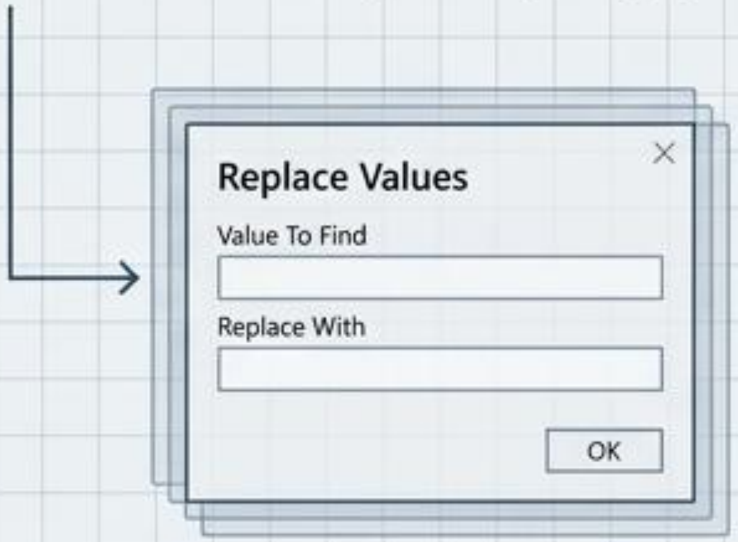
Treatment Protocol:
To replace, explicitly type the word 'null' (lowercase) into the 'Value To Find' field.



Blank

An empty text string.

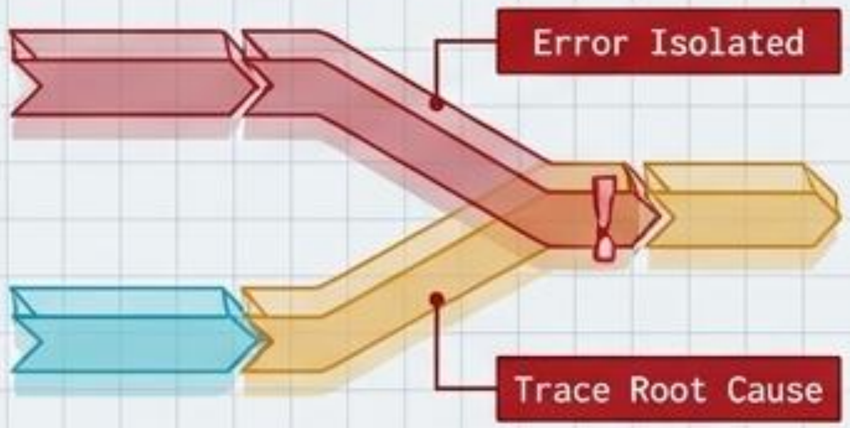
Treatment Protocol:
To replace, leave the 'Value To Find' field completely empty.



Error

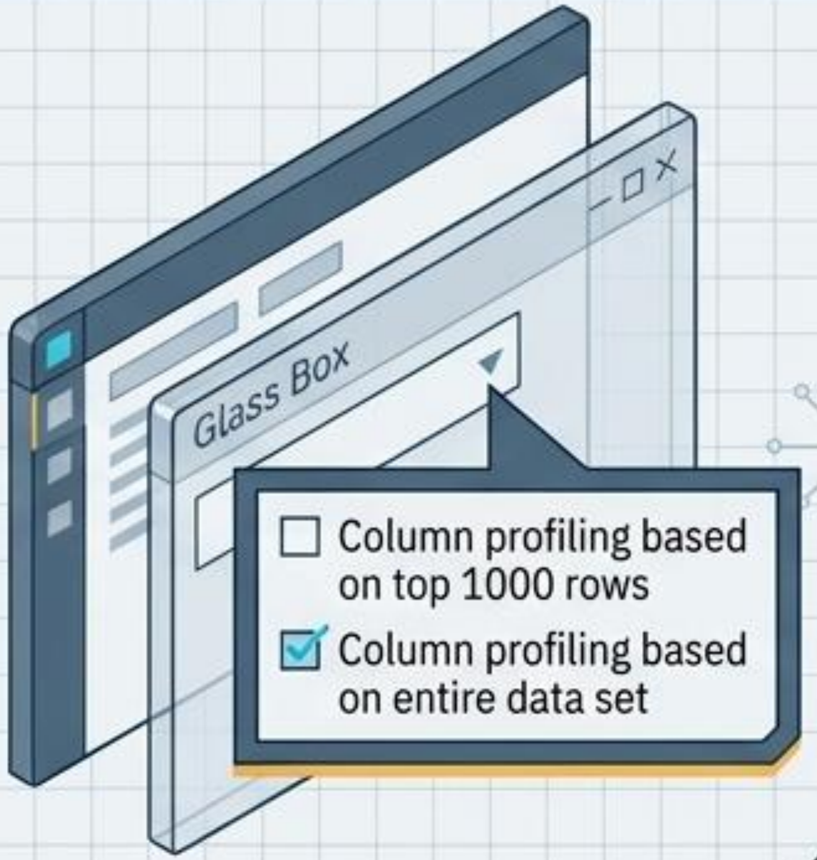
Data mismatch or failure.

Treatment Protocol:
Requires deep isolation and root-cause analysis rather than simple replacement.



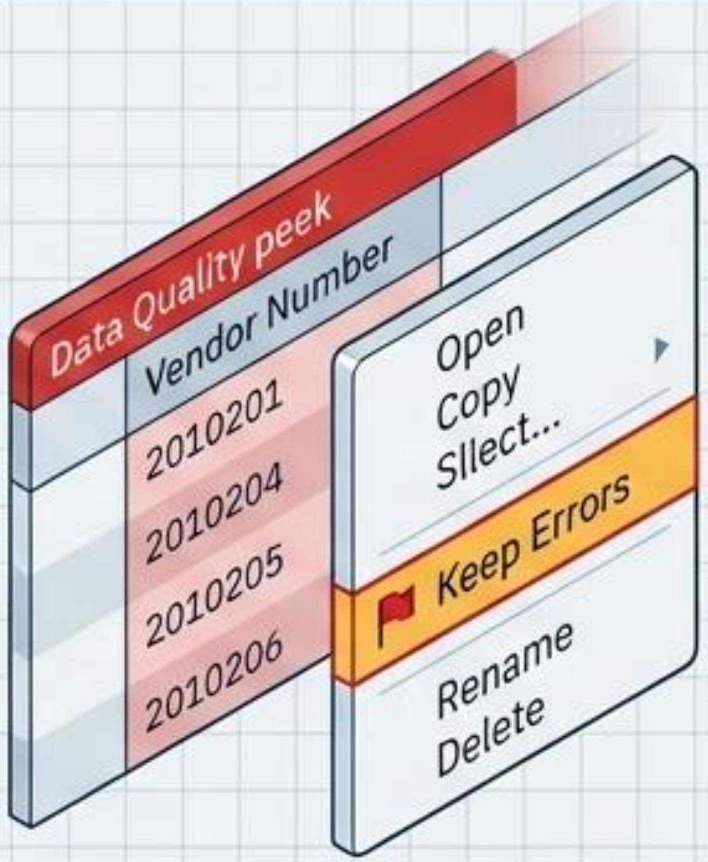
Isolating Deep-System Errors

Step 1: Expand the Net



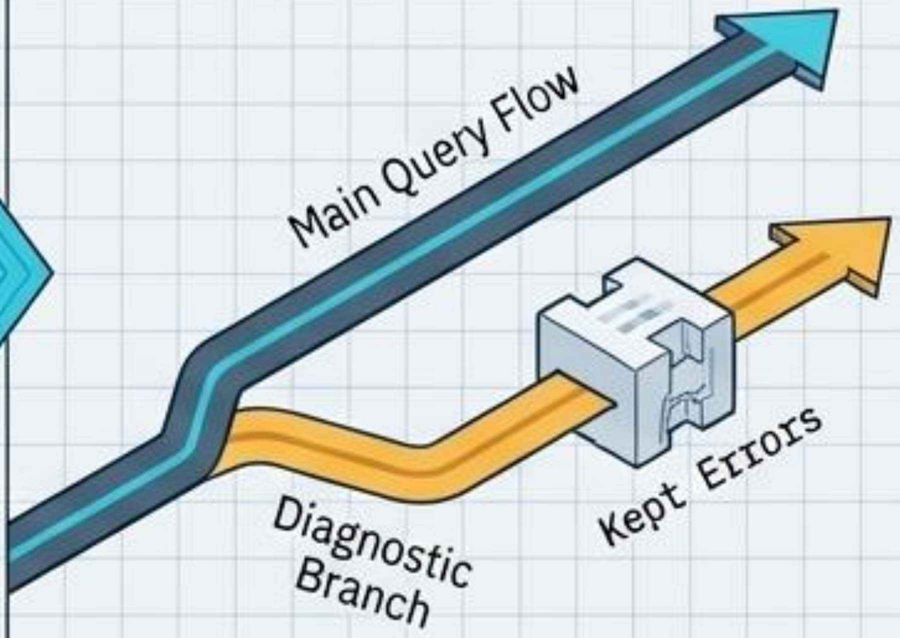
Change profiling from 'Top 1000 rows' to 'Entire data set' to reveal hidden errors far down the table.

Step 2: Isolate the Flaw



Select 'Keep Errors' to filter the view down exclusively to the broken rows.

Step 3: Branch & Diagnose



Duplicate the query before keeping errors to create a safe diagnostic branch without destroying the original data flow.

Phase 4: Precision vs. Scale in Data Typing

Rule: Fixed Decimal is not the equivalent of Decimal. The wrong data type is never a net-neutral decision.

Fixed Decimal

SPECIFICATION

Stores numbers at exactly four digits of precision after the decimal point.

USE CASE

Financial data and currency where rounding errors are completely unacceptable.

Decimal

SPECIFICATION

Stores up to 15 total digits (before and after the decimal place combined).

Danger: If a number has 13 digits before the decimal, it forces only 2 digits after, resulting in **irreversible loss of precision.**

The Storage Cost Hierarchy

Data types directly dictate the memory footprint of the final model.
Downgrade columns to their 'cheapest' viable format.

Text (String)

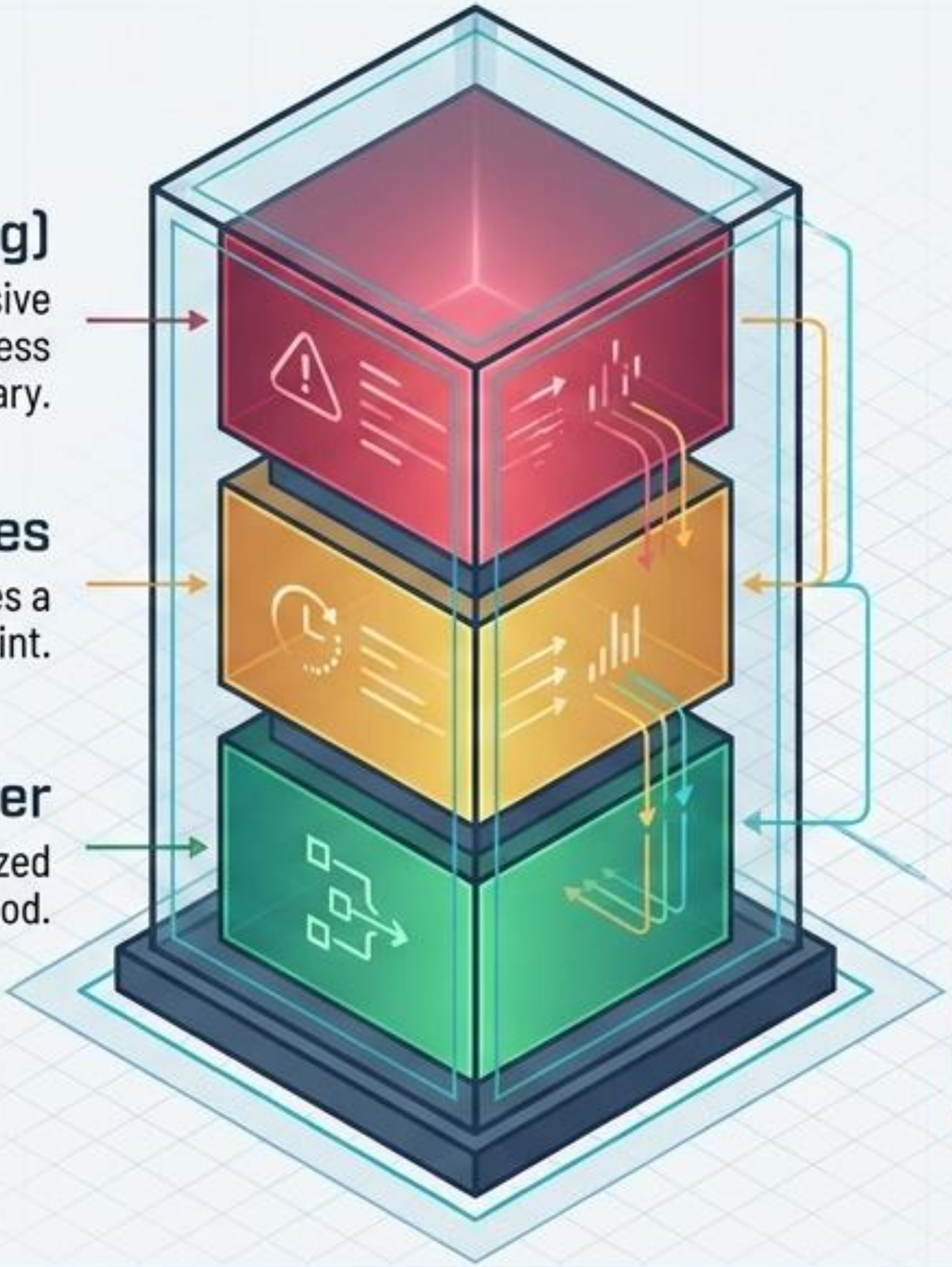
The most computationally expensive storage method. Avoid unless absolutely necessary.

Decimals & Dates

High utility but carries a moderate storage footprint.

Whole Number

The cheapest, most highly optimized and efficient storage method.

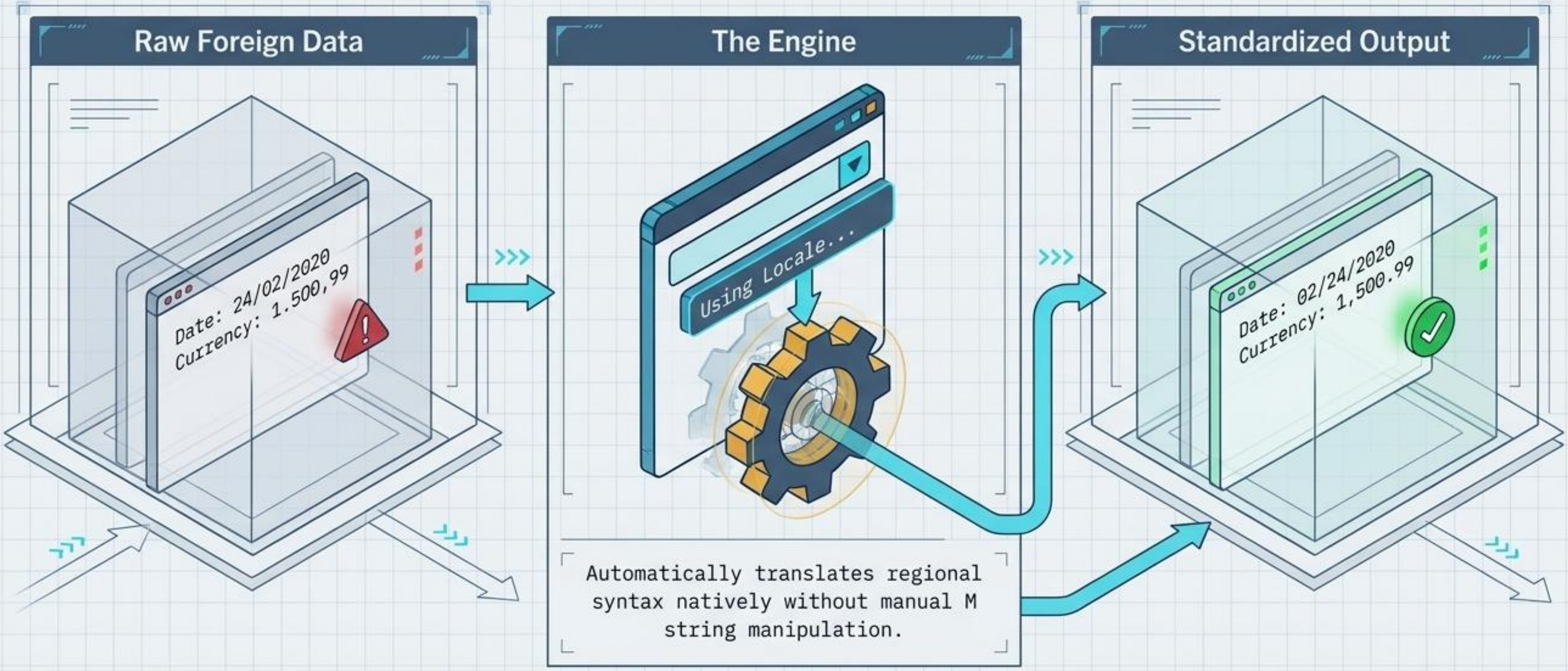


Glass Box

Real-World Impact: Changing a high-cardinality column from Text to Whole Number can reduce overall data model size by up to 30%.

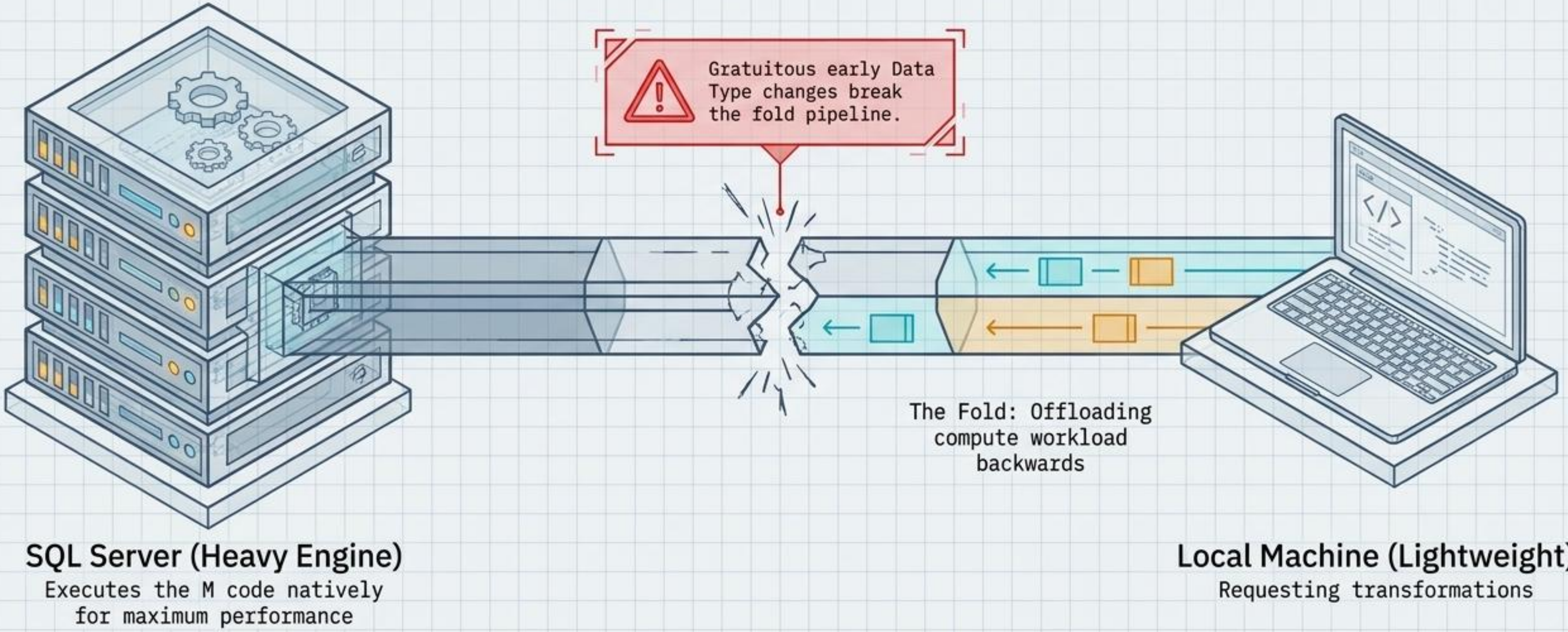
Automating Format Conversion via Locale

When adapting data for international consumers, never manually split or manipulate date strings.



Phase 5: Optimization via Query Folding

Whenever possible, the PQE "folds" transformation steps back to the source server engine to process them faster.



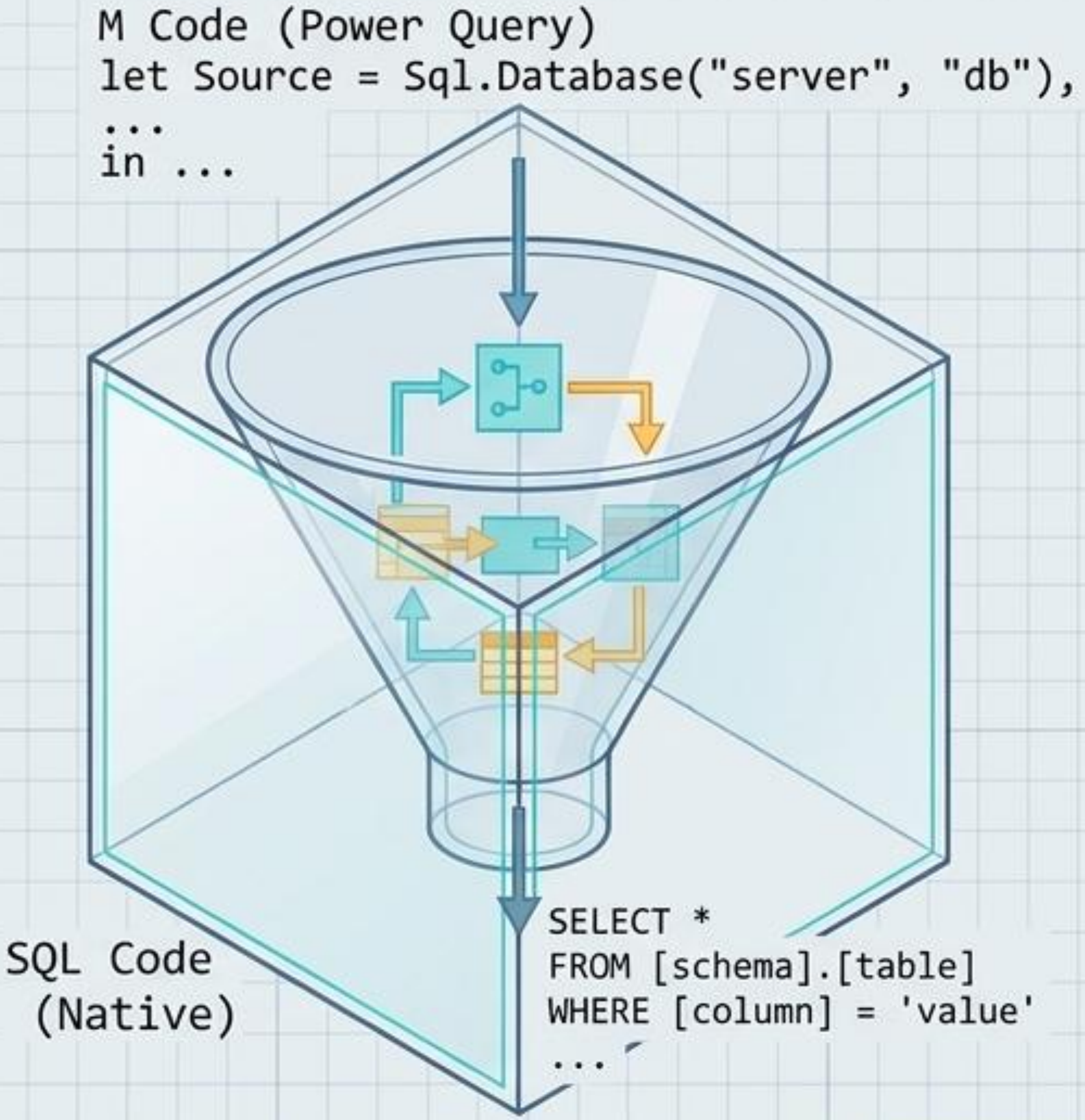
The Query Folding Audit

APPLIED STEPS

- Source
- Navigation
- Changed Type
- Filtered Rows
- Sorted Rows

View Native Query

The Test: Right-click any step. If View Native Query is grayed out, the folding pipeline has broken at or before that step (Flat files like CSVs never fold).



Final Assembly Line Rule: To maximize server pushback, Data Type changes must always be the **absolute final step** applied to your query. Delete any automatic type changes generated early in the sequence.